

Theorem 1. $\mathbb{R}$ is an ordered field with the Least Upper Bound Property

Proof. Begin with the proof that $\mathbb{R}$ is an ordered field.

For $\mathbb{R}$ to be an ordered field, it must be a field, it must be an ordered set, and it must satisfy the following:

(i) For all a, b , and c in $\mathbb{R}$, if $a < b$, then $a + c < b + c$.

(ii) If $a \leq b$ and $c \geq 0$, then $ac \leq bc$.

OEΔ